50 MW (AC) SOLAR PARK AT SUTIAKHALI, GOURIPUR, MYMENSINGH

Document Prepared by China International Engineering Consulting Corporation (Hong Kong) Limited

Project Title	50 MW (AC) SOLAR PARK AT SUTIAKHALI, GOURIPUR, MYMENSINGH
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1 PROJECT DETAILS

1.1 Summary Description of the Project

50 MW (AC) SOLAR PARK AT SUTIAKHALI, GOURIPUR, MYMENSINGH (hereinafter referred to as the proposed project) is to utilize the renewable solar energy for electricity generation through the construction of a solar photovoltaic (PV) power plant. The proposed project is located a 170-acre plot of land approximately 7km South East from the town centre of Mymensingh City, Bangladesh. The electricity generated from the project will be supplied to Bangladesh National Power Grid, which is owned and operated by Power Grid Company of Bangladesh Ltd. (PGCB). The proposed project is invested and developed by HDFC SinPower Limited. The estimated annual average net electricity generation supplied to the grid is 92,192 MWh and the annual full-load operation time amount to 1,844 h per year during the operation period. The estimated emission reduction is 55,302 tCO_2e annually and 387,113 tCO_2e of total emission reductions over the 7 years crediting.

The purpose of the proposed project is to generate solar power and deliver it to Bangladesh National Power Grid. For the proposed project,

(a) Prior to the start of implementation of the project activity, there is no solar photovoltaic power generation unit at the site of the proposed project, and the electricity was supplied by the Bangladesh National Power Grid which was dominated by fossil fuel-fired power plants.

(b) The project scenario is the implementation, installation and operation of a solar photovoltaic (PV) power plant. The total capacity of the project is proposed to be 60.177 MWp at DC level and 50MW at AC level, which will consist of 24 subsystems of around 2.5 MWp each. According to ACM0002 applied, the proposed project is a renewable electricity generating activity.

(c) The baseline scenario of the proposed project is the electricity supply of equal amount as the proposed project from the Bangladesh National Power Grid. The baseline scenario of the proposed project is the same as the scenario prior to the start of the implementation of the project activity.

(d) The project started construction on 22/09/2019 and the commercial operation start date is 04/11/2020.

The proposed project makes contribution to the local sustainable development as follows:

- The project will help reduce the greenhouse gas GHG emissions and the emission of SO_x, NO_x and dust by replacing the electricity generation from the fossil-fuel fired power plants of the grid.
- The conducting of the proposed project will create employment opportunities during the construction phase and operational period.
- The construction of the PV power plant will promote local economy by contributing to local government with more tax revenues through selling power generation.

1.2 Sectoral Scope and Project Type

The project activity falls under the following Sectoral scope and Project Type:

Sectoral Scope: 1 - Energy industries (renewable / non-renewable sources)

Project Type: Renewable energy project

The project is not a grouped project.

1.3 Project Eligibility

In line with the Section 2.1.1 of VCS Standard V4.3, the project eligibility is assessed below:

- It results in CO_2 emission reductions, one of the six Kyoto Protocol greenhouse gases.
- It consists in grid-connected electricity generation using solar power plant in Bangladesh, which is one of the Least Developed Countries (LDC) as per United Nations list of LDC[1].

Thus, the proposed project is eligible under the scope of the VCS program.

1.4 Project Design

The project has been designed to be a single installation of an activity, not a grouped project.

Eligibility Criteria

It is not applicable.

1.5 Project Proponent

Organization name	HDFC SinPower Limited
Contact person	Wang Hongqiang
Title	Plant Manager
Address	56 / 4-5 Anna South Bridge, Lake Circus Kalabagan, Dhaka-1205
Telephone	01760239773
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1.6 Other Entities Involved in the Project

Organization name	China International Engineering Consulting Corporation (Hong Kong) Limited

[1] https://unctad.org/topic/least-developed-countries/list

Role in the project	Project Consultant
Contact person	Zhang Hui
Title	Senior Project Manager
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1.7 Ownership

The project owner of the project is HDFC SinPower Limited. The approval of Environmental Impact Assessment (EIA), Project Approval, and the trade license of the project owner is evidence for legislative right.

1.8 Project Start Date

04/11/2020 (The commercial operation start date).

1.9 Project Crediting Period

This project adopts twice renewable for a total of 21 years, and the first crediting period is from 04/11/2020 to 03/11/2027.

1.10 Project Scale and Estimated GHG Emission Reductions or Removals

Project Scale	
Project	☑
Large project	

Year	Estimated GHG emission reductions or removals (tCO_2e)
04/11/2020~03/11/2021	56,076
04/11/2021~03/11/2022	55,818
04/11/2022~03/11/2023	55,560
04/11/2023~03/11/2024	55,302

04/11/2024~03/11/2025	55,044
04/11/2025~03/11/2026	54,786
04/11/2026~03/11/2027	54,527
Total estimated ERs	387,113
Total number of crediting years	7
Average annual ERs	55,302

1.11 Description of the Project Activity

This project would be a central generation-based grid connected PV system. As per FSR, the total capacity of this project is proposed to be a 60.177 MWp at DC level and 50MW at AC level, which will consist of 24 subsystems of around 2.5 MWp each. To make 2.5 MWp subsystem PV modules will be connected in series and then parallel. The DC power produced by PV modules is fed into grid-connected inverter. In the grid-connected inverter, the DC current is converted into AC current, flowing to 33kV step-up transformer. After that, 33kV voltage comes out and is going to be put into evacuation switch yard where the power will be step up into 132 kV and transported to nearby Kewatkhali substation. A set of monitoring systems planned to be used in this project, including: 33 kV power collection system, 33kV boost inverter systems and combiner boxes, PV modules, low-voltage power distribution equipment, DC systems. The output of the plant will be limited to 50MW (AC) through the monitoring system, generated power will be evacuated to nearby grid, finally to Bangladesh National Power Grid. The average load factor during the total operation period is calculated as 21.05% (=1844h/8760h).

Actually, the installed capacity of the project is 60.251 MWp at DC level and 50.05MW at AC level. The lifetime of the project is 20 years.

The plant's components characteristics are as follows:

Table 1: Characteristics of Solar PV Modules

Equipment	Manufacturer	Parameter	Unit	Technology standards	
Solar PV module	LONGI Solar Technology Co., Ltd.	Model	-	LR4-72HDB-435M	LR4-72HDB-430M
		Peak power	Wp	435	430
		Pieces	-	88,356	50,736
		Lifetime of Equipment	Year	20	
Inverter	Huawei Technologies Co., Ltd.	Model	-	SUN2000-185KTL-H1	
		Rated capacity	kW	175	
		Units	-	286	
		Power factor	-	0.99	

Prior to the implementation of the project, there is no solar power generation unit at the site of the proposed project, and the equivalent electricity supplied by the project was sourced from Bangladesh National Power Grid. By replacing the electricity generated from fossil fuel-fired power plants dominated Bangladesh National Power Grid, the project will achieve greenhouse gas (GHG) emission reductions by avoiding CO_2 emissions.

The baseline scenario of the proposed project is the same as the scenario prior to the start of the implementation of the project activity.

1.12 Project Location

The project is located at located a 170-acre plot of land approximately 7km South from the centre of Mymensingh City, Bangladesh. The geographical coordinates for the project site are 24°42'13.39"N (24.70372°) and 90°27'49.81" E (90.46384°).

The Figure 1-1a shows the location of Mymensingh City in Bangladesh and Figure 1-1b shows the location of the project site.

Figure 1-1a The location of Mymensingh City in Bangladesh

Figure 1-1b The location of the project site

1.13 Conditions Prior to Project Initiation

The project activity is the installation of a new grid-connected photovoltaic power project and the generated power will be exported to the Bangladesh National Power Grid. Electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources within Bangladesh National Power Grid.

The project is a renewable resource based on photovoltaic power project without GHG emissions during operation period. Therefore, it was confirmed that the project has not been implemented to generate GHG emissions for the purpose of their subsequent reduction, removal or destruction.

1.14 Compliance with Laws, Statutes and Other Regulatory Frameworks

The Project has received necessary approvals for development and commissioning of the Solar Power Project from the government of Bangladesh or the authorized agencies including:

1. E-Trade Licence by Bangladesh local government (Dhaka South City Corporation).
2. Grid access approval by Bangladesh Power Development Board.
3. Environmental Clearance & approval of EIA by Department of Environment (DoE) Bangladesh.

Renewable Energy policy (Renewable Energy Policy of Bangladesh -2008[2]) and the policies like Bangladesh Energy Regulatory Commission Act, 2003[3] and Bangladesh Electricity Act 2018[4] are framed primarily to encourage renewable energy power projects but does not influence the choice of fuel used for power generation. Moreover, there is no legal requirement on the choice of a particular technology or fuel for power generation. Implementation of Solar Power projects is therefore a voluntary activity and is not mandated by any legal mandates in Bangladesh.

1.15 Participation under Other GHG Programs

1.15.1 Projects Registered (or seeking registration) under Other GHG Program(s)

The project activity has not been registered, nor is it seeking registration under any other GHG program.

1.15.2 Projects Rejected by Other GHG Programs

The project has not been rejected by any other GHG program.

1.16 Other Forms of Credit

1.16.1 Emissions Trading Programs and Other Binding Limits

The project proponent is not part of any emission trading program. The net GHG emission reductions from the project will not be used for compliance with emission trading programs or to meet binding limits on GHG emissions. The project activity has not participated under any other GHG programs.

1.16.2 Other Forms of Environmental Credit

The project hasn't sought or received another form of environmental credits.

1.17 Sustainable Development Contributions

1.17.1 Sustainable Development Contributions Activity Description

The project activity includes implementation of 50 MW Grid-connected Solar Power Project in Bangladesh. Electricity generated from the project activity is supplied to Bangladesh Power Development Board, thus substitutes high carbon intensive grid electricity by clean and renewable based electricity. Installation of the project activity will enhance the current renewable energy share in the national grid as well as contribute towards reducing greenhouse gas emission

[2] http://www.sreda.gov.bd/d3pbs_uploads/files/policy_1_rep_english.pdf

[3] Annual Report, Bangladesh Power Development Board.

[4] https://www.bpdb.gov.bd/bpdb_new/d3pbs_uploads/files/11%20March%2019/Electricity%20Act%20(Gazette).pdf

which would have otherwise been emitted due to generation of electricity in the fossil fuel powered power projects. The project will also result in job creation both on temporary basis (during the construction stage) and permanent basis (during operation phase) including employability of youth.

SDG Goal	Initiatives	Expected contribution to sustainable development
SDG 7– Ensure access to affordable, reliable, sustainable and modern energy for all	The project activity will result in generation of electricity by using solar energy which will in turn be fed to the national grid of Bangladesh.	The project will increase substantially the share of renewable energy in the energy mix of Bangladesh National Grid from renewable based electricity generated by the project activity and fed to the grid.
SDG 8 - Promote sustained, inclusive and sustainable economic growth, full and productive employment and descent work for all	Implementation (including construction and operation) of the project activity required involvement and engagement of human resources.	The project activity has resulted in employability both on temporary basis (during construction) and permanent basis (during operation) including employment of youth.
SDG 13 - Take urgent action to combat climate change and its impacts	Project activity generates renewable energy-based electricity and mitigates the CO_2 emissions which would have been generated from the fossil fuel-based power plants.	Feeding of renewable energy-based electricity to the grid will replace generation of electricity from fossil fuel-based power projects that is currently supplying electricity to the grid and reduce equivalent greenhouse gas emission.

1.17.2 Sustainable Development Contributions Activity Monitoring

The Project activity includes implementation of a grid connected solar power project that feds in the electricity generated to the grid and thereby increase the share of renewable energy in existing national grid. Feeding of renewable based electricity to the grid will result in refraining of generation of electricity in the fossil fuel -based power plant that currently supplies power to the grid resulting in reduction of equivalent amount of greenhouse gas emission. The project activity will also result in both temporary and permanent employability.

Table 1: Sustainable Development Contributions

Row number	SDG Target	SDG Indicator	Net Impact on SDG Indicator	Current Project Contributions	Contributions Over Project Lifetime
Sequential row number	*SDG Target number*	*Number and text of SDG indicator or, if no official SDG indicator is applicable, user-defined indicator*	*Indicate the project's contribution to the SDG Indicator (implemented activities to increase or decrease)*	*Brief description of the quantifiable impact of the project's activities related to the SDG indicator, during the monitoring period.*	*Brief description of the cumulative quantifiable impact of the project's activities related to the SDG indicator, over the project lifetime.*
1)	7.2	7.2.1 Renewable energy share in the total final energy consumption	The project being a renewable energy-based power generation unit feeding power to the grid will enhance the renewable energy share in the final energy consumption.	The project has contributed to the SDG goals by supplying 171,859.680 MWh of renewable electricity to the grid until the end of 08/2022.	From 04/11/2020 onwards, 171,859.680 MWh of electricity from renewable sources has been supplied to the power grid.
2)	8.5	8.5.1 Average hourly earnings of employees, by sex, age, occupation and persons with disabilities	The project activity will result in temporary and primary employment opportunity during the construction and operation phase.	The project has contributed to the SDG goals by creating employment opportunities for about 336 persons including 300 persons during the construction period and 36 persons during the operation period.	From 04/11/2020 onwards, the project has provided 336 employment positions for local residents.
3)	13.0	Tonnes of greenhouse gas emissions avoided or removed	Implemented activities to result in avoidance of greenhouse Gas emission by refraining generation of electricity in fossil fuel -based power plant connected to the grid.	Generation of 171,859.680 MWh of renewable electricity that is fed to the grid the project activity will result in avoidance of 99,987 tCO_2e during the monitoring period (04/11/2020 to 31/08/2022)	Prevented the annual release of 99,987 tonnes of carbon into the atmosphere from 04/11/2022 up to the end of this monitoring period (31/08/2022).

1.18 Additional Information Relevant to the Project

Leakage Management

Not appliable as the project is not a AFOLU project.

Commercially Sensitive Information

No commercially sensitive information has been excluded from the public version of the project description.

Further Information

No situation is identified that have a bearing on the eligibility of the project, the net GHG emission reductions or removals, or the quantification of the project's net GHG emission reductions or removals.

2 SAFEGUARDS

2.1 No Net Harm

No potential negative environmental or socio -economic impacts have been identified for the project and the impacts identified in the project are of minor significance. Project Proponent has conducted detailed Environmental Impact Assessment (EIA) to assess the Environment, Social and economic impacts of the solar PV based power project activity during its construction & operation phases to ensure the appropriate mitigation action be planned to be in place if there is any negative impact.

The EIA clearly indicated that the environmental impact of the project activity is of minor significance. The EIA outlines the social and economic benefits (as identified) linked to employment generation, local procurement, encouragement of local enterprise development and skill development within the communities from the project activity. The project in its entirety is likely to bring prosperity and development into the region and pave the way for greater decarbonisation of national grid.

In addition, the project activity is contributing towards reducing the GHG emission by replacing equivalent amount of electricity generated from fossil fuel fired power plants and contributing towards energy self-sufficiency.

2.2 Local Stakeholder Consultation

Local Stakeholder Consultation during the project preparation stage:

To investigate the impacts on local ecological environment and social environment, local stakeholder consultation was carried out among the potential stakeholders as a part of the EIA study in March 2017. The survey was arranged through a questionnaire, which was designed to be easily filled in with the following sections:

a) Do you think the construction of this project will help improve the surrounding environment?

b) Do you think the construction of this project will promote the development of local economy?

c) Do you think this project will increase job opportunities for local residents?

d) In your opinion, what are the possible responsible impacts of the construction of this project?

e) Do you support the construction of this emission reduction project?

Consultations was conducted during the EIA with the objective of obtaining view of the project stakeholders about the planned project including their feedbacks, and concerns through individual interviews and focused group discussions.

As a part of the consultation process, the project representatives explained all stakeholders about the type of development through the project and the technology used for power generation towards enhancing stakeholders' awareness about the project. The stakeholders were also made aware about the employment opportunities and other benefits likely to be rendered through this project and the importance of setting up solar PV power plant, not only to tap the cleaner sources for energy generation but most to meet the ever-increasing energy demand of state and its contribution towards fighting against climate change.

30 copies questionnaires were distributed in March 2017. The recovery rate is 100%. The comments, from the government, environmental protection bureau, local farmers and other relevant stakeholders, were collected. There are no negative comments received for the project according to every questionnaire survey. The survey shows that the project received strong support from local people. Most of the respondents have concerned about the environmental impacts. The EIA ensured that, any grievance, suggestions or general feedback are accepted and addressed in a timely manner and incorporated in the project where applicable and relevant.

Local Stakeholder Consultation during the project implementation stage:

The project owner has established the Stakeholder Engagement Plan (SEP)[5]. The mechanism of communication with local stakeholders has been constructed for the implementation stage of the project.

Approaches of Stakeholders consultation mainly includes:

1. Tracking information and activities;

[5] Stakeholder Engagement Plan (SEP), HDFC SinPower Ltd., Document No.: HDFC SEP-PMC-171226, 26 December 2017

2. Written communication;
3. Information dissemination;
4. Individual consultations;
5. Focused group discussions (FGDs);
6. Household level consultations;
7. Public consultations.

Communications with local stakeholders are being carried out at periodic intervals. Key implementation schedules or changes of the project will be communicated to the local authority, who will inform the neighbourhood committee and the local residents, the comments and suggestions from residents will be collected by the local authority meanwhile. And the local government agencies and competent authorities will conduct spot checks on the implementation of the project from time to time and give suggestions on the involved rectification problems. In line with VCS requirements all the processes have been implemented to receive comments from local stakeholders as well as communicate with them at periodic intervals.

In this monitoring period from 04/11/2020 to 31/08/2022, the project owner carried out the focused group discussions to collect the relevant comments and suggestions on 26/12/2020 and 20/11/2021. The main topics focused on investigating the social and environmental impacts of the project. The comments, from the local government authority, grid company, local residents and other relevant stakeholders, were collected. The communities and the community representatives in and around study area expressed the view that the project had a positive impact There are no negative comments received for the project according to the survey.

2.3 Environmental Impact

No negative environmental impacts have been identified from the project based on the EIA undertaken by the project proponent. The project proponent has obtained environmental clearance for the project activity.

2.4 Public Comments

To be obtained.

2.5 AFOLU-Specific Safeguards

This section is not applicable as the project is a non - AFOLU project.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

Following approved baseline & monitoring methodology is applied:

Methodology:

ACM0002 "Grid-connected electricity generation from renewable sources" (Version 20.0)

Reference: https://cdm.unfccc.int/methodologies/DB/XP2LKUSA61DKUQC0PIWPGWDN8ED5PG

The applied tools:

Tool 01- "Tool for the demonstration and assessment of additionality" - （Version 7.0.0, EB 70 annex 08)

Reference: https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-01-v7.0.0.pdf

Tool 07 – "Tool to calculate the emission factor for an electricity system" -(Version 07.0, EB 100 annex 4)

Reference: https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-07-v7.0.pdf

Tool 24 – "Common practice" -(Version 03.1, EB 84 annex 7)

Reference: https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-24-v1.pdf

Tool 27- "Investment analysis"- (Version 11.0, EB 112 annex 2)

Reference: https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-27-v11.0.pdf

3.2 Applicability of Methodology

The approved methodology ACM0002 (version 20.0) is applicable to the project activity and the project meets the applicability of the applied methodology as follows:

Clauses	Requirements of the ACM0002	Scenario of the project	Conclusion
1	This methodology is applicable to grid-connected renewable energy power generation project activities that: a) Install a Greenfield power plant; b) Involve a capacity addition to (an) existing plant(s); c) Involve a retrofit of (an) existing operating plants/units; d) Involve a rehabilitation of (an) existing plant(s)/unit(s); or e) Involve a replacement of (an) existing plant(s)/unit(s).).	The project is a greenfield grid-connected renewable power generation project.	Applicable

Clauses	Requirements of the ACM0002	Scenario of the project	Conclusion
2	The methodology is applicable under the following conditions: (a) The project activity may include renewable energy power plant/unit of one of the following types: hydro power plant/unit with or without reservoir, wind power plant/unit, geothermal power plant/unit, solar power plant/unit, wave power plant/unit or tidal power plant/unit; (b) In the case of capacity additions, retrofits, rehabilitations or replacements (except for wind, solar, wave or tidal power capacity addition projects the existing plant/unit started commercial operation prior to the start of a minimum historical reference period of five years, used for the calculation of baseline emissions and defined in the baseline emission section, and no capacity expansion, retrofit, or rehabilitation of the plant/unit has been undertaken between the start of this minimum historical reference period and the implementation of the project activity.	The project is a newly built solar power project and the project activity involves the installation of the solar power plant.	Applicable
3	In case of hydro power plants, one of the following conditions must apply: (a) The project activity is implemented in existing single or multiple reservoirs, with no change in the volume of any of the reservoirs; or (b) The project activity is implemented in existing single or multiple reservoirs, where the volume of the reservoir(s) is increased and the power density calculated using equation (7), is greater than 4 W/m^2; or (c) The project activity results in new single or multiple reservoirs and the power density, calculated using equation (7), is greater than 4 W/m^2; or (d) The project activity is an integrated hydro power project involving multiple reservoirs, where the power density for any of the reservoirs, calculated using equation (7), is lower than or equal to 4	Not applicable, the project is not a hydro power plant, so this applicability condition does not need to be considered.	NA

Clauses	Requirements of the ACM0002	Scenario of the project	Conclusion
	W/m^2, all of the following conditions shall apply: (i) The power density calculated using the total installed capacity of the integrated project, as per equation (8), is greater than 4 W/m^2; (ii) Water flow between reservoirs is not used by any other hydropower unit which is not a part of the project activity; (iii) Installed capacity of the power plant(s) with power density lower than or equal to 4 W/m^2 shall be: a. Lower than or equal to 15 MW; and b. Less than 10 per cent of the total installed capacity of integrated hydro power project.		
4	In the case of integrated hydro power projects, project proponent shall: Demonstrate that water flow from upstream power plants/units spill directly to the downstream reservoir and that collectively constitute to the generation capacity of the integrated hydro power project; or Provide an analysis of the water balance covering the water fed to power units, with all possible combinations of reservoirs and without the construction of reservoirs. The purpose of water balance is to demonstrate the requirement of specific combination of reservoirs constructed under CDM project activity for the optimization of power output. This demonstration has to be carried out in the specific scenario of water availability in different seasons to optimize the water flow at the inlet of power units. Therefore, this water balance will take into account seasonal flows from river, tributaries (if any), and rainfall for minimum five years prior to implementation of CDM project activity.	Not applicable, the project is not a hydro power plant, so this applicability condition does not need to be considered.	NA
5	The methodology is not applicable to: a) Project activities that involve switching from fossil fuels to renewable energy sources at the site of the project	The project does not involve switching from fossil-fuels to renewable energy sources at the site of the project activity	NA

Clauses	Requirements of the ACM0002	Scenario of the project	Conclusion
	activity, since in this case the baseline may be the continued use of fossil fuels at the site; b) Biomass fired power plants/units.	and the project is not a biomass-fired power project.	
6	In the case of retrofits, rehabilitations, replacements, or capacity additions, this methodology is only applicable if the most plausible baseline scenario, as a result of the identification of baseline scenario, is "the continuation of the current situation, that is to use the power generation equipment that was already in use prior to the implementation of the project activity and undertaking business as usual maintenance".	Not applicable, the project is a newly built solar power project.	NA

In addition, the project meets the applicability conditions of the applied tools applied in the PD as follows:

Tool	Criteria	Applicability	Conclusion
Tool for the demonstration and assessment of additionality (Version 07.0.0)	The use of the "Tool for the demonstration and assessment of additionality" is not mandatory for project participants when proposing new methodologies. Project participants may propose alternative methods to demonstrate additionality for consideration by the Executive Board. They may also submit revisions to approved methodologies using the additionality tool.	The methodology selected for the proposed project requires the use of this tool.	Applicable
	Once the additionally tool is included in an approved methodology, its application by project participants using this methodology is mandatory.	The methodology applied in this proposed project requires the use of this tool.	Applicable
Tool to calculate the emission factor for an electricity system (Version 07.0)	This tool may be applied to estimate the OM, BM and/or CM when calculating baseline emissions for a project activity that substitutes grid electricity that is where a project activity supplies electricity to a grid or a project activity that results in savings of electricity that would have been provided by the grid (e.g.	The project is the installation of a solar power plant supplying electricity to the grid.	Applicable

Tool	Criteria	Applicability	Conclusion
	demand-side energy efficiency projects).		
	In case of CDM projects the tool is not applicable if the project electricity system is located partially or totally in an Annex I country.	The project electricity system is located in a non-Annex I country.	NA
Investment analysis (Version 11.0)	This methodological tool is applicable to project activities that apply the methodological tool “Tool for the demonstration and assessment of additionality”, the methodological tool “Combined tool to identify the baseline scenario and demonstrate additionality”, the guidelines “Non-binding best practice examples to demonstrate additionality for SSC project activities”, or baseline and monitoring methodologies that use the investment analysis for the demonstration of additionality and/or the identification of the baseline scenario	The project applies the methodological tool “Tool for the demonstration and assessment of additionality”.	Applicable
	In case the applied approved baseline and monitoring methodology contains requirements for the investment analysis that are different from those described in this methodological tool, the requirements contained in the methodology shall prevail.	The methodology ACM0002 (Version 20.0) applied in this project requires the use of this tool to demonstrate the investment analysis of this project.	Applicable
Common practice (Version 03.1)	This methodological tool is applicable to project activities that apply the methodological tool “Tool for the demonstration and assessment of additionality”, the methodological tool “Combined tool to identify the baseline scenario and demonstrate additionality”, or baseline and monitoring methodologies that use the common practice test for the demonstration of additionality.	This project applies the methodological tool “Tool for the demonstration and assessment of additionality”.	Applicable

Tool	Criteria	Applicability	Conclusion
	In case the applied approved baseline and monitoring methodology defines approaches for the conduction of the common practice test that are different from those described in this methodological tool, the requirements contained in the methodology shall prevail.	The methodology ACM0002 (Version 20.0) applied in this project requires the use of this tool to demonstrate the common practice of this project.	Applicable

3.3 Project Boundary

The spatial extent of the project boundary includes the project power plant / unit and all power plants / units connected physically to the electricity system that the project power plant is connected to. The project boundary therefore includes the project site where the power plant has been installed, associated power evacuation infrastructure, energy metering points, switch yards and other civil constructions and the connected Bangladesh National Power Grid.

The project boundary includes CO_2e as greenhouse gas (GHG) that would have emanated from the operation of grid connected fossil fuel -based power plants in the absence of the project activity.

The table below provides an overview of the emissions sources included or excluded from the project boundary for determination of baseline and project emissions.

Source		Gas	Included?	Justification/Explanation
Baseline	CO_2 emissions from electricity generation in fossil fuel fired power plants that are displaced due to the project activity	CO_2	Yes	Main emission source
		CH_4	No	Minor emission source
		N_2O	No	Minor emission source
Project	The project activity	CO_2	No	As a zero-emission grid connected Solar power project no GHG emissions will result from the project activity in operation phase.
		CH_4	No	As a zero-emission grid connected Solar power project no GHG emissions

Source		Gas	Included?	Justification/Explanation
				will result from the project activity in operation phase.
		N_2O	No	As a zero-emission grid connected Solar power project no GHG emissions will result from the project activity in operation phase.

3.4 Baseline Scenario

As per the approved consolidated Methodology ACM0002 (Version 20.0), baseline scenario would be defined as: “If the project activity is the installation of a Greenfield power plant, the baseline scenario is electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in the “Tool to calculate the emission factor for an electricity system (Version 07.0)”.

The project activity involves setting up of solar power projects to use the renewable energy (solar energy) resource to produce electricity and supply it to the Bangladesh national grid (Bangladesh National Power Grid). In the absence of the project activity, the equivalent amount of power would have generated by the operation of existing grid connected power plants (fossil fuel dominated) and by the addition of new generation sources. Hence, the baseline for the project activity is the equivalent amount of power of the Bangladesh national grid that are displaced due to the project activity.

During the implementation of the project activity, the relevant national and/or sectoral policies, regulations and circumstances are taken into account.

- Implementation of solar PV based power generation unit for electricity generation is not mandatory under any law in Bangladesh, the project activity is thus a voluntary action.
- Despite the increase in renewable energy sources (including solar energy) in power sector, still more than ninety -eight percent of installed power generation capacity is based on fossil -fuel based energy sources, hence the electricity grid is fed by electricity generated predominantly in fossil-fuel based power plants.

Construction of Solar PV based power plants are exempted from Environmental Impact Assessment (EIA). The Project has however received approval from Bangladesh Electricity Regulatory Commission and Department of Environment (DoE) Bangladesh.

3.5 Additionality

In accordance with the methodology ACM0002 (Version 20.0), the additionality of the project activity is demonstrated and assessed using the latest version of the "Tool for the demonstration and assessment of additionality" (version 7.0. 0). The tool provides a step-wise approach to demonstrate and assess the additionality of a project as below:

(a) Step 0 Demonstration whether the proposed project activity is the first-of-its-kind;

(b) Step 1 Identification of alternatives to the project activity;

(c) Step 2 Investment analysis;

(d) Step 3 Barriers analysis; and

(e) Step 4 Common practice analysis.

The step-wise approach to establish additionality of the project activity has been followed, details of which are provided in the following paragraphs:

Step 0: Demonstration whether the proposed project activity is the first -of-its-kind;

The project activity is a large-scale solar power project undertaken in Bangladesh. This is not the first such project to be installed in the country and therefore project activity does not meet this criterion.

Step 1: Identification of alternatives to the project activity consistent with current laws and regulations

Define realistic and credible alternatives to the project activity(s) through the following Sub-steps.

Sub-step 1a: Define alternatives to the project activity

Alternatives available to the project participant or similar project developers that provide outputs or services comparable with the project activity are identified below:

Alternative (a): The proposed project activity undertaken without being registered as a VER project activity.

The option of proposed project activity being undertaken without being registered under VERRA is in compliance with all applicable legal and regulatory requirements and can be a part of the baseline. However, this is not a plausible option as deployment of the project activity faces significant financial barriers established in the subsequent section. The electricity generated from the project activity would have been fed to the Bangladesh National Grid, which has a number of power plants including existing and upcoming ones which would be used in the absence of project activity. In the absence of project activity, power from the same grid will continue to provide for the requirement of the country.

Alternative (b) Continuation of the current situation (no project activity or other alternatives undertaken).

In such case the project owner would have continued without investment in project activity and continued with its usual business activities. In such case grid would continue with the fossil fuel based power projects and this would result in GHG emissions. Hence, existing power plant and the new capacity add-on from a fossil fuel-based power plant is appropriate, realistic & credible baseline alternative for the project activity.

Outcome of Sub-step 1a: Identified realistic and credible alternative scenario(s) to the project activity

Of the two alternatives outlined above, the first alternative is not possible as project activity is not feasible without carbon credit benefits and second alternative is the baseline scenario for the project activity as per methodology. Therefore, continuation of the current situation is the likely alternative to the project activity.

The project being a green field project activity the baseline scenario in line with the methodology is "the baseline scenario is electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid -connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in the "Tool to calculate the emission factor for an electricity system", Version 7.0.

Sub-step 1b: Consistency with mandatory laws and regulations

Both alternatives (a) and (b) are realistic and credible alternatives to the project which are consistent with mandatory laws and regulations.

Outcome of Sub-step 1b: Identified realistic and credible alternative scenario(s) to the project activity that are in compliance with mandatory legislation and regulations taking into account the enforcement in the region or country and EB decisions on national and/or sectoral policies and regulations.

Hence, both the alternatives identified above are found to comply with the mandatory laws and regulations taking into account the enforcement of the legislations in the region/country and EB decisions on national and/or sec toral policies and regulations. However, Alternative (b) "Continuation of the current situation" has been selected as the appropriate baseline alternative for this project activity in line with methodology.

Step 2: Investment analysis

Investment analysis is carried out to determine on whether the proposed project activity is economically or financially less attractive than at least one other alternative, identified in step 1, without the revenue from the sale of emission reductions credits. This is demonstrated in in line with the Tools for sections as per "Investment Analysis" (version 11.0).

Sub-step 2a: Determine appropriate analysis method

According to TOOL01 (version 7.0.0), if the project activity and the alternatives identified in Step 1 generate no financial or economic benefits other than carbon related income, then apply the simple cost analysis (Option I). Otherwise, use the investment comparison analysis (Option II) or the benchmark analysis (Option III).

For this project, the alternative to the project activity is the supply of electricity from a grid which is not considered as an investment by project participant, so the benchmark analysis (Option III) will be used.

Sub-step 2b: Option III. Apply benchmark analysis

The investment analysis using Benchmark analysis approach (Option III) has been chosen. Further, this method illustrates that the evaluation of the project by the project owner was carried out before the decision to undertake the project was taken and management approval had been granted.

Choice of Financial Indicator:

According to the "Tool for demonstration and assessment of Additionality" Version 07.0.0, the financial indicator can be based either on (1) project IRR or (2) equity IRR. There is no general preference between the approaches (1) or (2). The benchmark chosen for analysis shall be fully consistent with the choice of approach. Therefore, in accordance with the guidance, the relevant financial indicator for project activity has been chosen as post tax Equity IRR.

Default Value Benchmark:

The cost of equity is determined by selecting the values provided in the Appendix, i.e. Default values for cost of equity (expected return on equity) is presented below:

Appendix of the CDM TOOL 27: "Investment Analysis" （Version 11.0） specifies default value of expected return on equity in real terms for Energy Industries (Group 1) in Bangladesh = 11.72%.

The Required return on equity (benchmark) was computed in the following manner:

Nominal Benchmark = {(1+Real Benchmark) *(1+Inflation rate)} -1

Where:

Default value for Group 1 project category Real Benchmark = 11.72% (as per Appendix of CDM TOOL 27: "Investment Analysis"

Inflation rate forecast for Bangladesh：The average forecasted inflation rate for the host country published by the Asian Development Bank for the next four years (only four years' data is available) after the start of the project activity has been used. The average inflation rate forecast is 5.80%[6].

The nominal benchmark is calculated to be 18.20% as per the above formula.

[6] https://www.adb.org/countries/bangladesh/economy

Sub-step 2c: Calculation and comparison of financial indicators (only applicable to Options II and III)

The IRR and Benchmark analysis is calculated in excel spreadsheet and is also summarized below. Based on result of IRR excel spreadsheets, equity IRR is assessed to be less than Benchmark. This substantiates that the investment is not financially attractive (Equity IRR for the project activity is less than the Benchmark). Thus, it can be easily concluded that project activity is additional & is not business as usual scenario.

The main financial parameters and Equity IRR of the project are shown in Table 2 and Table 3 below.

Table 2: Financial Parameters

Item	Unit	Value	Data Source
Capacity	MW	50	FSR
Operation Hour in first year	h	1928	FSR
Total Investment	10^4USD	9,806.15	FSR
Static investment	10^4USD	9,594.15	FSR
Tariff (excluding tax)	USD/kWh	0.17	FSR
First year attenuation rate	%	3.00%	FSR
Attenuation rate of each year since then	%	0.50%	FSR
Scheduled repayment period	year	12	FSR
Long-term loan interest rate	%	7.07%	FSR
Annual net Electricity Supplied	MWh	92,192	FSR
Operation Life	Year	20	FSR
Interest during construction period	10^4USD	212	FSR
Personnel quota standard	Person	11	FSR
Average Labor Wage	10^4USD/year	1.93	FSR
Comprehensive welfare expenses	-	70%	FSR
Insurance rate of fixed assets	-	0.250%	FSR
Material cost	USD/kW	0.99	FSR
Other cost	USD/kW	5.68	FSR
Surplus accumulation fund percentage	-	10%	FSR
Working capital	USD/kW	6.40	FSR
Capital ratio	-	30%	FSR
Liquidity capital ratio	-	30%	FSR
Depreciation period	year	20	FSR

Residual rate	-	5%	FSR
Rate of Repair Fee during the first two years	-	0%	FSR
Rate of Repair Fee from 3rd year to 5th year	-	0.6%	FSR
Rate of Repair Fee from the 6th year to the 20th year	-	1.1%	FSR

Table 3: Project IRR with and without carbon credit

Items	Equity IRR (EIRR)
Without ER revenue	11.11%
With ER revenue	11.91%

Sub-step 2d: Sensitivity analysis

The objective of sensitivity analysis is to show whether the conclusion regarding the financial attractiveness is robust to reasonable variations in the critical assumptions. The investment analysis provides a valid argument in favour of additionality only it consistently supports (for a realistic range of assumptions) the conclusion that the project activity is unlikely to be the most financially attractive or is unlikely to be financially attractive.

As per the Guidelines on the assessment of investment analysis (version 05.0), only the variables constituting more than 20% of either total project costs or total project revenues and having significant impact on the sensitivity analysis should be subjected to reasonable variation. Thus, for the project, the following financial parameters were taken as uncertain factors for sensitive analysis of financial attractiveness:

- Total static investment
- Annual operation cost
- Annual electricity output
- Electricity tariff

Therefore, considering the reasonable variations in the critical assumptions, the sensitivity analysis is made regarding the 4 key factors of total static investment, electricity tariff, annual operation cost and annual electricity delivered to the grid, as shown in Table 4, Figure 3.

Table 4: Sensitivity analysis of EIRR for the project

Sensitivity Analysis						
Items	-10.00%	-5.00%	0.00%	5.00%	10.00%	Critical variation
Tariff (Excl. VAT)	7.18%	9.10%	11.11%	13.21%	15.39%	16.16%-
Annual electricity	7.18%	9.10%	11.11%	13.21%	15.39%	16.16%-

Total static investment	15.51%	13.15%	11.11%	9.34%	7.79%	14.89%
Annual operation cost	11.87%	11.49%	11.11%	10.73%	10.34%	100.90%

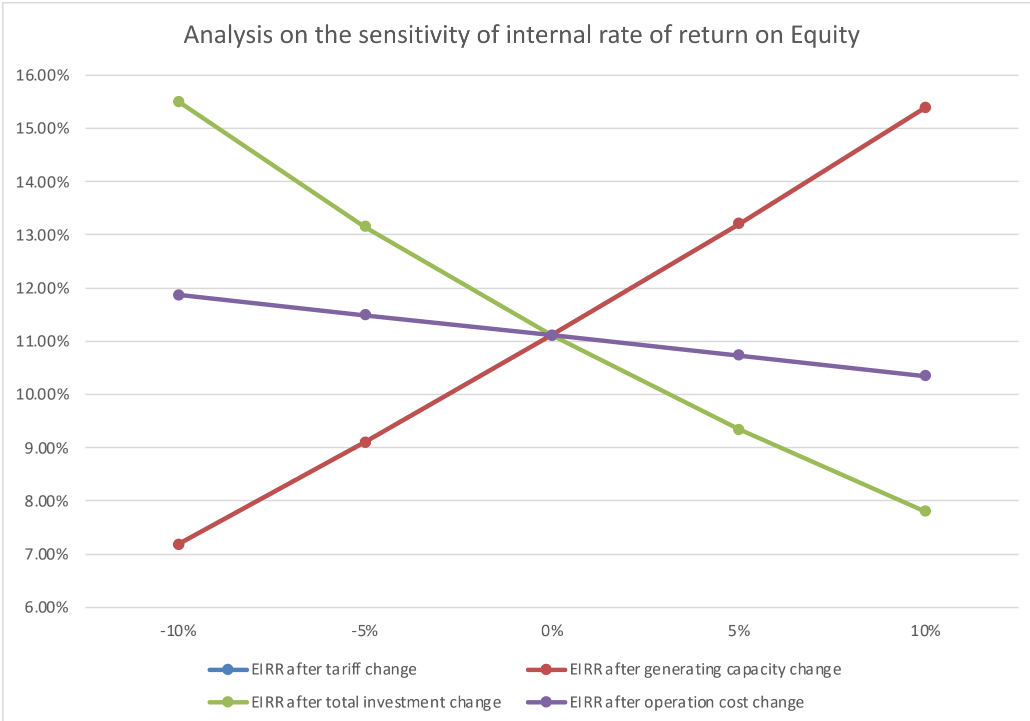


Figure 3: Sensitivity analysis for the project Equity

Total static Investment

The Project EIRR would reach the benchmark rate of 18.20% when the total static investment decreases by 14.89%. According to the investment details by the project owner, the actual investment of the project (11,000 10^4USD) is 12.17% higher than the estimated total investment (9,806.15 10^4USD). Therefore, it is impossible that the total static investment can be reduced by 14.89% to make the EIRR to cross the benchmark.

Annual operation cost

The Project EIRR would reach the benchmark when the annual operation cost decreases by 100.90% which means the operation cost will be negative value to make the EIRR reaching benchmark. It is obviously impossible.

Annual electricity generation and electricity tariff (Electricity revenue)

The Project EIRR would reach the benchmark when the electricity revenue increased by 16.16%. According to FSR prepared by the third party, the supplied electricity is estimated based on the project capacity and the project load factor. It is unlikely to fluctuate up to 16.16% all over the operation period to reach the benchmark. The tariff fixed in the power purchase agreement of the project is 0.17 USD/kWh (excluding tax) same as the value estimated in the FSR. Therefore, it is unlikely that the electricity revenue would increase by 16.16%.

Step 3: Barrier analysis

The proposed project does not employ the barrier analysis. Then proceed to Step 4.

Step 4: Common practice analysis

The section below provides the analysis as per step 4 of the TOOL01 "Tool for the demonstration and assessment of additionality", version 7.0.0 and according to TOOL24 "Common Practice", version 03.1.

As the project applies power generation based on renewable energy which is one of the measures(s) listed in the definitions section above, proceed to Sub-step 4a.

Sub-step 4a: The proposed project activity(ies) applies measure(s) that are listed in the definitions section above Common practice analysis is carried out as per CDM TOOL24 "Common practice" version 03.1.

Sub-Step 4a-1: calculate applicable capacity or output range as +/-50% of the total design capacity or output of the proposed project activity

The install capacity of the project is 50 MW, so the applicable capacity range as +/-50% of the total design capacity is 25 ~ 75 MW.

Sub-Step 4a-2: identify similar projects which fulfil all of the following conditions:

a. The projects are located in the applicable geographical area;

As the project is located Bangladesh, therefore, the applicable geographic area is Bangladesh where the project located. The projects in the geographical area of Bangladesh have been chosen for analysis.

b. The projects apply the same measure as the proposed project activity;

The applicable measure is power generation based on renewable energy, same as the project.

c. The projects use the same energy source/fuel and feedstock as the proposed project activity, if a technology switch measure is implemented by the proposed project activity;

The energy source is the renewable solar power, same as the project.

d. The plants in which the projects are implemented produce goods or services with comparable quality, properties and applications areas (e.g. clinker) as the proposed project plant;

The applicable project is to produce electricity, same as the project.

e. The capacity or output of the projects is within the applicable capacity or output range calculated in Step 1;

As defined in Step 4a-1, the applicable capacity range is from 25 MW to 75 MW.

f. The projects started commercial operation before the project document (PD) is published for global stakeholder consultation or before the start date of proposed project activity, whichever is earlier for the proposed project activity.

The start date of the proposed project is 04/11/2020 which is earlier than the PD published, so the applicable commercial operation starting date is before 04/11/2020.

Sub-Step 4a-3: within the projects identified in Step 2, identify those that are neither registered project activities, project activities submitted for registration, nor project activities undergoing validation. Note their number, N_{all}.

According to the available statistics (such as annual reports of Bangladesh Power Development Board) and relevant project information on CDM, VCS and GS website, no similar projects can be identified within the applicable range. Therefore, $N_{all} = 0$.

Sub-Step 4a-4: within similar projects identified in Step 3, identify those that apply technologies that are different to the technology applied in the proposed project activity. Note their number N_{diff}.

Since $N_{all} = 0$, also $N_{diff} = 0$, $N_{all} = N_{diff}$.

Sub-Step 4a-5: calculate factor $F=1-N_{diff}/N_{all}$ representing the share of similar projects (penetration rate of the measure/technology) using a measure/technology similar to the measure/technology used in the proposed project activity that deliver the same output or capacity as the proposed project activity.

As stated before, $N_{all} = N_{diff}$, $F = 1 - N_{diff}/N_{all} = 1 - 1 = 0<0.2$, therefore the project activity is NOT a “common practice” within a sector in the applicable geographical area, according to the guideline

3.6 Methodology Deviations

The project did not apply any methodology deviations.

4 ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

According to the ACM0002 (Version 20.0), baseline emissions include only CO_2 emissions from electricity generation in fossil fuel fired power plants that are displaced due to the project activity, calculated as follows:

$$BE_y = EG_{PJ,y} \times EG_{grid,CM,y} \quad (1)$$

Where:

BE_y	= Baseline emission in year y (tCO_2/yr)
$EG_{PJ,y}$	= Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the VCS project activity in year y (MWh/yr)
$EF_{grid,CM,y}$	= Combined margin CO_2 emission factor for grid connected power generation in year y calculated using the latest version of the "Tool to calculate the emission factor for an electricity system"

The project activity is a Greenfield solar power plant, then:

$$EG_{PJ,y} = EG_{facility,y} \quad (2)$$

Where:

$EG_{PJ,y}$	= Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the VCS project activity in year y (MWh/yr)
$EG_{facility,y}$	= Quantity of net electricity generation supplied by the project plant/unit to the grid in year y (MWh/yr)

As per methodology, combined grid emission factor is estimated as per the "Tool to calculate the emission factor for an electricity system", version 07.0 is presented below:

Step 1. Identify the relevant electricity system

According to the "Tool to calculate the emission factor for an electricity system", version 07.0, project participants may delineate the project electricity system using any of the following options:

Option 1. A delineation of the project electricity system and connected electricity systems published by the DNA or the group of the DNAs of the host country(ies), In case a delineation is provided by a group of DNAs, the same delineation should be used by all the project participants applying the tool in these countries;

Option 2. A delineation of the project electricity system defined by the dispatch area of the dispatch centre responsible for scheduling and dispatching electricity generated by the project activity.

Where the dispatch area is controlled by more than one dispatch centre, i.e. layered dispatch area, the higher-level area shall be used as a delineation of the project electricity system (e.g. where regional dispatch centres are required to comply with dispatch orders of the national dispatch centre then area controlled by the national dispatch centre shall be used);

Option 3. A delineation of the project electricity system defined by more than one independent dispatch areas, e.g. multi-national power pools.

Keeping the options into consideration, Department of Environment (DoE), Ministry of the Environment and Forest (MoEF) of Bangladesh, has considered the national grid of Bangladesh while developing national baseline on power and energy sector[7]. Power Grid Company of Bangladesh is the sole grid operating agency and wheels the entire grid connected power. It is a government owned company operating in Bangladesh and a subsidiary of Bangladesh Power Development Board.

Step 2. Choose whether to include off-grid power plants in the project electricity system (optional)

Project participants may choose between the following two options to calculate the operating margin and build margin emission factor:

Option I: Only grid power plants are included in the calculation.

Option II: Both grid power plants and off-grid power plants are included in the calculation.

The Project Participant has chosen only grid power plants in the calculation.

Step 3. Select a method to determine the operating margin (OM)

The calculation of the operating margin emission factor ($EF_{grid,OM,y}$) is based on the following methods:

(a) Simple OM, or

(b) Simple adjusted OM, or

(c) Dispatch Data Analysis OM, or

(d) Average OM.

The data required to calculate Simple adjusted OM and Dispatch data analysis OM is not possible due to lack of availability of data to project developers. The choice of other two options for calculating operating margin emission factor depends on generation of electricity from low -cost/ must-run sources. In the context of the methodology low cost/must run resources typically include hydro, geothermal, wind, low -cost biomass, nuclear and solar generation.

In accordance to the tool the simple OM method can only be used if any one of the following requirements is satisfied:

1. Low-cost/must-run resources constitute less than 50 per cent of total grid generation (excluding electricity generated by off -grid power plants) in: 1) average of the five most recent years, and the average of the five most recent years shall be determined by using one of the approaches described under the tools; or 2) based on long-term averages for hydroelectricity production (minimum time frame of 15 years).

[7] Clean Development Mechanism (CDM) baseline, Department of Environment, Ministry of Environment

2. The average amount of load (MW) supplied by low -cost/must-run resources in a grid in the most recent three year.

Share of low -cost/must-run (% of Net Generation)

Year[8]	2015-2016	2016-2017	2017-2018	2018-2019	2019-2020
Total Generation (GWh)	52,193	57,276	62,678	70,534	71,419
Power generation by Hydro (GWh)	962.20	982.04	1,024.31	724.65	825.19
Power generation by other renewable energy (GWh)	0.13	0.09	3.79	39.00	62.00
Share of low -cost/must-run, %	1.84%	1.71%	1.64%	1.08%	1.24%

Data source: Annual Report of Bangladesh Power Development Board

The above data clearly shows that the percentage of total grid generation by low -cost/ must-run plants (on the basis of average of five most recent years) for the Bangladesh grid is less than 50 % of the total generation. Thus, the Average OM method cannot be applied, as low cost/must run resources constitute less than 50% of total grid generation. The simple OM emission factor is calculated as the generation-weighted average CO_2 emissions per unit net electricity generation (tCO_2/MWh) of all generating power plants serving the system, not including low -cost/must-run power plants/units.

For the simple OM, the simple adjusted OM and the average OM, the emissions factor can be calculated using either of the two following data vintages:

(a) Ex-ante option: if the ex-ante option is chosen, the emission factor is determined once at the validation stage, thus no monitoring and recalculation of the emissions factor during the crediting period is required.

OR

(b) Ex-post option: if the ex-post option is chosen, the emission factor is determined for the year in which the project activity displaces grid electricity, requiring the emissions factor to be updated annually during monitoring.

Project proponent has chosen ex-ante option for calculation of Simple OM emission factor using a 3-year generation-weighted average, based on the most recent data available at the time of submission of the PD to the VVB for project validation. OM determined at project validation stage will be the same throughout the crediting period. There will be no requirement to monitor & recalculate the emission factor during the crediting period.

Step 4: Calculate the operating margin emission factor according to the selected method

[8] The year is corresponding to the fiscal year of Bangladesh Power Development Board (BPDB) in Annual Reports.

The simple OM emission factor is calculated as the generation-weighted average CO_2 emissions per unit net electricity generation (t CO_2/MWh) of all generating power plants serving the system, not including low-cost/must-run power plants/units.

The simple OM may be calculated by one of the following two options:

(a) Option A: Based on the net electricity generation and a CO_2 emission factor of each power unit; or

(b) Option B: Based on the total net electricity generation of all power plants serving the system and the fuel types and total fuel consumption of the project electricity system. Option B can only be used if: (i) The necessary data for Option A is not available; and (ii) Only nuclear and renewable power generation are considered as low cost/must -run power sources and the quantity of electricity supplied to the grid by these sources is known; and (iii) Off-grid power plants are not included in the calculation (i.e. if Option I has been chosen in Step 2).

Option A: Calculation based on average efficiency and electricity generation of each plant

Under this option, the simple OM emission factor is calculated based on the net electricity generation of each power unit and an emission factor for each power unit, as follows:

$$EF_{grid,OMsimple,y} = \frac{\sum_m EG_{m,y} \times EF_{EL,m,y}}{\sum_m EG_{m,y}} \quad (3)$$

Where:

$EF_{grid,OMsimple,y}$	=	Simple operating margin CO_2 emission factor in year y (tCO_2 /MWh)
$EG_{m,y}$	=	Net quantity of electricity generated and delivered to the grid by power unit m in the year y (MWh)
$EF_{EL,m,y}$	=	CO_2 emission factor of power unit m in year y (tCO_2 /MWh)
m	=	All power units serving the grid in year y except low-cost/must-run power units
y	=	The relevant year as per data vintage chosen in Step 3

Determination of $EF_{EL,m,y}$

Option A1 - If for a power unit m data on fuel consumption and electricity generation is available, the emission factor ($EF_{EL,m,y}$) should be determined as follows:

$$EF_{EL,m,y} = \frac{\sum_i FC_{i,m,y} \times NCV_{i,y} \times EF_{CO2,i,y}}{EG_{m,y}} \quad (4)$$

Where:

$EF_{EL,m,y}$	=	CO_2 emission factor of power unit m in year y (tCO_2 /MWh)
$FC_{i,m,y}$	=	Amount of fuel type *i* consumed by power unit m in year y (Mass or volume unit)
$NCV_{i,y}$	=	Net calorific value (energy content) of fuel type i in year y (GJ/mass or volume unit)
$EF_{CO2,i,y}$	=	CO_2 emission factor of fuel type i in year y (t CO_2/GJ)
$EG_{m,y}$	=	Net quantity of electricity generated and delivered to the grid by power unit m in year y (MWh)
m	=	All power units serving the grid in year y except low-cost/must-run power units
i	=	All fuel types combusted in power unit m in year y
y	=	The relevant year as per data vintage chosen in Step 3

Electricity imports are treated as one power plant, as per the tool guidance.

Option A2 - If for a power unit m only data on electricity generation and the fuel types used is available, the emission factor should be determined based on the CO2 emission factor of the fuel type used and the efficiency of the power unit, as follows:

$$EF_{EL,m,y} = \frac{EF_{co2,m,i,y} \times 3.6}{\eta_{m,y}} \quad (5)$$

Where:

$EF_{EL,m,y}$	=	CO_2 emission factor of power unit m in year y (tCO_2/MWh)
$EF_{CO2,m,i,y}$	=	Average CO_2 emission factor of fuel type i used in power unit m in year y (t CO_2/GJ)
$\eta_{m,y}$	=	Average net energy conversion efficiency of power unit m in year y (ratio)
m	=	All power units serving the grid in year y except low-cost/must-run power units
y	=	The relevant year as per data vintage chosen in Step 3
3.6	=	Conversion factor (GJ/MWh)

The Simple OM calculation and the data used for the calculation is provided in the EF excel sheet. The resulting values of this calculation are as follows:

The operating margin emission factor ($EF_{grid,OM,y}$) has been calculated using a 3-year data vintage:

Calculation of the Operating Margin (OM) emission factor of Bangladesh National Grid			
Year[9]	2017-2018	2018-2019	2019-2020
Net electricity delivered to the grid (GWh, include imports)	62,678	70,533	71,419
Annual simple OM (tCO_2/MWh)	0.6228	0.6037	0.5743
$EF_{grid,OM,y}$ (tCO_2/MWh)	0.5992		

Step 5: Calculate the build margin (BM) emission factor

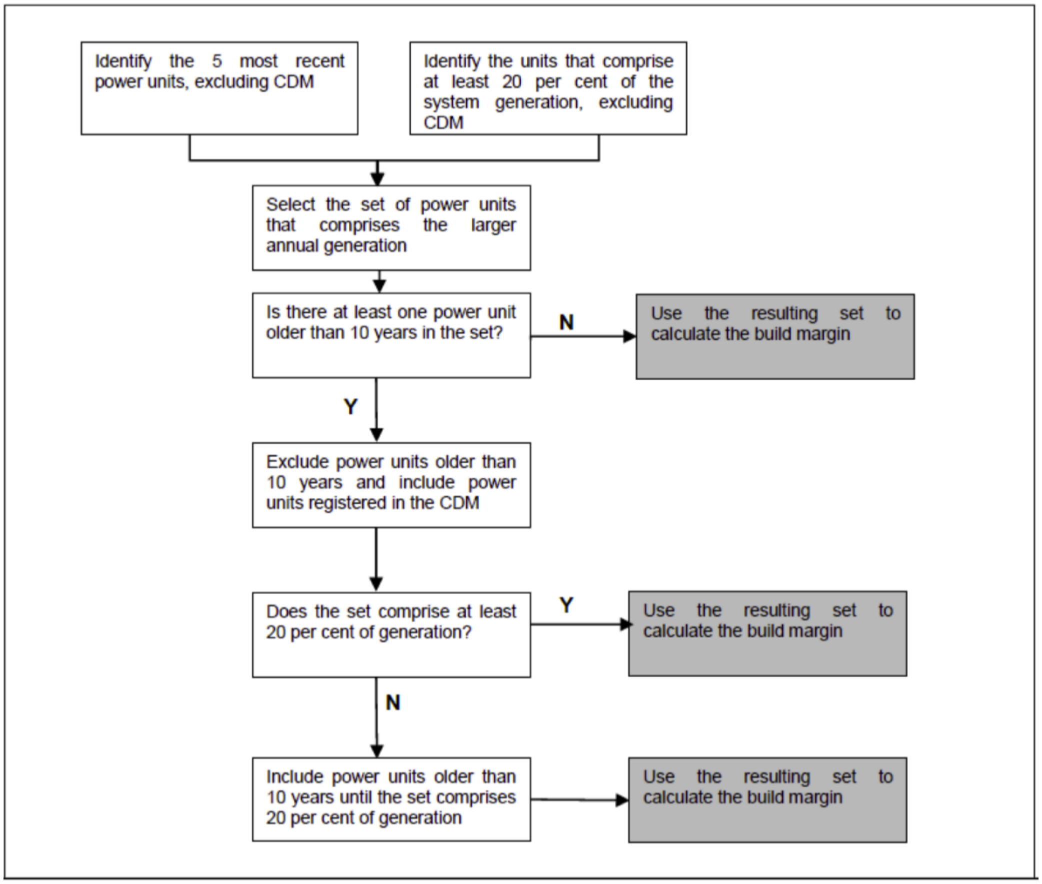
As per Methodological tool: "Tool to calculate the emission factor for an electricity system", version 07.0 para 72: In terms of vintage of data, project participants can choose between one of the following two options:

(a) Option 1 - for the first crediting period, calculate the build margin emission factor ex ante based on the most recent information available on units already built for sample group at the time of PD submission to the VVB for project validation. For the second crediting period, the build margin emission factor should be updated based on the most recent information available on units already built at the time of submission of the request for renewal of the crediting period to the VVB. For the third crediting period, the build margin emission factor calculated for the second crediting period should be used. This option does not require monitoring the emission factor during the crediting period.

(b) Option 2 - For the first crediting period, the build margin emission factor shall be updated annually, ex post, including those units built up to the year of registration of the project activity or, if information up to the year of registration is not yet available, including those units built up to the latest year for which information is available. For the second crediting period, the build margin emissions factor shall be calculated ex ante, as described in Option 1 above. For the third crediting period, the build margin emission factor calculated for the second crediting period should be used.

Option 1 as described above is chosen by PP to calculate the build margin emission factor for the project activity. BM is calculated ex -ante based on the most recent information available at the time of submission of PD and is fixed for the entire crediting period.

[9] The year is corresponding to the fiscal year of Bangladesh Power Development Board (BPDB) in Annual Reports.



The build margin emissions factor is the generation -weighted average emission factor (tCO_2/MWh) of all power units m during the most recent year y for which electricity generation data is available, calculated as follows:

$$EF_{grid,BM,y} = \frac{\sum_m EG_{m,y} \times EF_{EL,m,y}}{\sum_m EG_{m,y}} \quad (6)$$

Where:

$EF_{grid,BM,y}$	=	Build margin CO_2 emission factor in year y (tCO_2/MWh)
$EG_{m,y}$	=	Net quantity of electricity generated and delivered to the grid by power unit m in year y (MWh)
m	=	Power units included in the build margin
y	=	Most recent historical year for which electricity generation data is available

The Year 2019-2020 is selected as the most recent historical year for which electricity generation data is available. The build margin emissions factor is calculated as the generation -weighted average emission factor (tCO_2/MWh) of all power units m during the Year 2019-2020 without adjusted with imports. $EF_{grid,BM,y}$ is calculated to be 0.5300 tCO_2e/MWh. The calculation process is presented in EXCEL spreadsheet.

Step 6: Calculate the combined margin emissions factor

The calculation of the combined margin (CM) emission factor ($EF_{grid,CM,y}$) is based on one of the following methods:

(a) Weighted average CM; or

(b) Simplified CM.

PP has chosen option (a) i.e weighted average CM to calculate the combined margin emission factor for the project activity.

The combined margin emissions factor is calculated as follows:

$$EF_{grid,CM,y} = EF_{grid,OM,y} \times W_{OM} + EF_{grid,BM,y} \times W_{BM} \quad (7)$$

Where:

$EF_{grid,BM,y}$	=	Build margin CO_2 emission factor in year y (tCO_2/MWh)
$EF_{grid,OM,y}$	=	Operating margin CO2 emission factor in year y (t CO2/MWh)
W_{OM}	=	Weighting of operating margin emissions factor (per cent)
W_{BM}	=	Weighting of build margin emissions factor (per cent)

The combined margin emissions factor $EF_{grid,CM,y}$ should be calculated as the weighted average of the Operating Margin emission factor ($EF_{grid,OM,y}$) and the Build Margin emission factor ($EF_{grid,BM,y}$), where W_{OM}= 0.75 and W_{BM} = 0.25 for solar power generation project (owing to their intermittent and non-dispatchable nature) for the first crediting period and for subsequent crediting periods. The ($EF_{grid,OM,y}$) and ($EF_{grid,BM,y}$) are calculated as described in Step 4 and 5.

$EF_{grid,CM,y}$ = 0.5992 tCO_2e/MWh *0.75+0.5300 tCO_2e/MWh *0.25 = 0.5818 (tCO_2e/MWh)

4.2 Project Emissions

According to ACM0002 (version 20.0), the project is a solar farm, belongs to renewable energy activity, hence PE_y =0 tCO_2e/MWh.

4.3 Leakage

According to ACM0002 (version 20.0), no leakage is considered for this project.

4.4 Estimated Net GHG Emission Reductions and Removals

The annual emission reductions ER_y for the project activity are calculated as the baseline emissions minus the project emissions. Being the project of a greenhouse gas GHG zero-emission activity, the final GHG emission reductions are calculated as follows:

$$ER_y = BE_y - PE_y \quad (8)$$

Where:

ER_y = Emission reductions in year y (tCO_2)

BE_y = Baseline Emissions in year y (tCO_2)

PE_y = Project emissions in year y (tCO_2)

And the baseline emissions (BE_y in tCO_2) are the product of the baseline emissions factor ($EF_{grid,CM,y}$ in tCO_2/MWh) times the net electricity supplied by the project activity to the grid ($EG_{facility}$ in MWh). The calculation formula is as follows:

$$BE_y = EG_{facility,y} \times EG_{grid,CM,y} \quad (9)$$

The summary of ex ante estimates of emission reductions is shown as follows:

Year	Estimated baseline emissions or removals (tCO_2e)	Estimated project emissions or removals (tCO_2e)	Estimated leakage emissions (tCO_2e)	Estimated net GHG emission reductions or removals (tCO_2e)
Year 1	56,076	0	0	56,076
Year 2	55,818	0	0	55,818
Year 3	55,560	0	0	55,560
Year 4	55,302	0	0	55,302
Year 5	55,044	0	0	55,044
Year 6	54,786	0	0	54,786
Year 7	54,527	0	0	54,527
Total	387,113	0	0	387,113

5 MONITORING

5.1 Data and Parameters Available at Validation

Data / Parameter	$EF_{grid,OM,y}$
Data unit	tCO_2e/MWh
Description	Simple operating margin CO_2 emission factor in year y

Source of data	Calculated follow the "Tool to calculate the emission factor for an electricity system" (Version 07.0) Data source: Annual Reports of Bangladesh Power Development Board www.bpdb.gov.bd/site/page/c4161d54-5b85-4917-a8d2-68a2d1b26dd4/মাসিক-বাসষক-প্রসিববদন
Value applied:	0.5992
Justification of choice of data or description of measurement methods and procedures applied	Calculated follow the "Tool to calculate the emission factor for an electricity system" (Version 07.0)
Purpose of Data	Calculation of baseline emissions
Comments	N/A

Data / Parameter	$EF_{grid,BM,y}$
Data unit	tCO_2e/MWh
Description	Build margin CO_2 emission factor in year y
Source of data	Calculated follow the "Tool to calculate the emission factor for an electricity system" (Version 07.0) Data source: Annual Reports of Bangladesh Power Development Board www.bpdb.gov.bd/site/page/c4161d54-5b85-4917-a8d2-68a2d1b26dd4/মাসিক-বাসষক-প্রসিববদন
Value applied:	0.5300
Justification of choice of data or description of measurement methods and procedures applied	Calculated follow the "Tool to calculate the emission factor for an electricity system" (Version 07.0)
Purpose of Data	Calculation of baseline emissions
Comments	N/A

Data / Parameter	W_{OM}

Data unit	-
Description	Weighting of operating margin emissions factor
Source of data	"Tool to calculate the emission factor for an electricity system" (version 07.0)
Value applied:	0.75
Justification of choice of data or description of measurement methods and procedures applied	Follow the "Tool to calculate the emission factor for an electricity system" (Version 07.0)
Purpose of Data	Calculation of $EF_{grid,CM,y}$
Comments	N/A

Data / Parameter	W_{BM}
Data unit	-
Description	Weighting of building margin emissions factor
Source of data	"Tool to calculate the emission factor for an electricity system" (version 07.0)
Value applied:	0.25
Justification of choice of data or description of measurement methods and procedures applied	Follow the "Tool to calculate the emission factor for an electricity system" (Version 07.0)
Purpose of Data	Calculation of $EF_{grid,CM,y}$
Comments	N/A

Data / Parameter	$EF_{grid,CM,y}$
Data unit	tCO_2e/MWh
Description	Combined margin CO_2 emission factor in year y
Source of data	Calculated follow the "Tool to calculate the emission factor for an electricity system" (Version 07.0)
Value applied:	0.5818

Justification of choice of data or description of measurement methods and procedures applied	Calculated follow the “Tool to calculate the emission factor for an electricity system” (Version 07.0)
Purpose of Data	Calculation of baseline emissions
Comments	N/A

5.2 Data and Parameters Monitored

Data / Parameter	$EG_{facility,y}$
Data unit	MWh
Description	Quantity of net electricity generation supplied by the project to the Grid in year y.
Source of data	Calculated
Description of measurement methods and procedures applied	The net electricity supplied to the Grid by the project will be calculated by the electricity exported to the grid ($EG_{export,y}$) and the electricity imported from the grid ($EG_{import,y}$) by the project through the following formula: $EG_{facility,y} = EG_{export,y} - EG_{import,y}$ The electricity exported and imported from the grid monitored and measured by Main Meter and backup Meter installed at the 132kV side of the unit transformer within the project site.
Frequency of monitoring/recording	Monitoring continuously and recording monthly
Value applied:	92,192 MWh (The estimated annual average value during the whole operation period in FSR), and the parameter will be monitored in the monitoring period.
Monitoring equipment	Two bidirectional electricity meters M1(main meter) and M2 (backup meter) with accuracy of no lower than 0.5 are installed at the project site.
QA/QC procedures applied	The data will be cross checked with sales receipts. The electricity meter(s) will be periodically checked and maintained. Calibration will be carried out periodically to ensure normal operation.
Purpose of data	Calculation of baseline emissions
Calculation method	Not applicable

Comments	-

5.3 Monitoring Plan

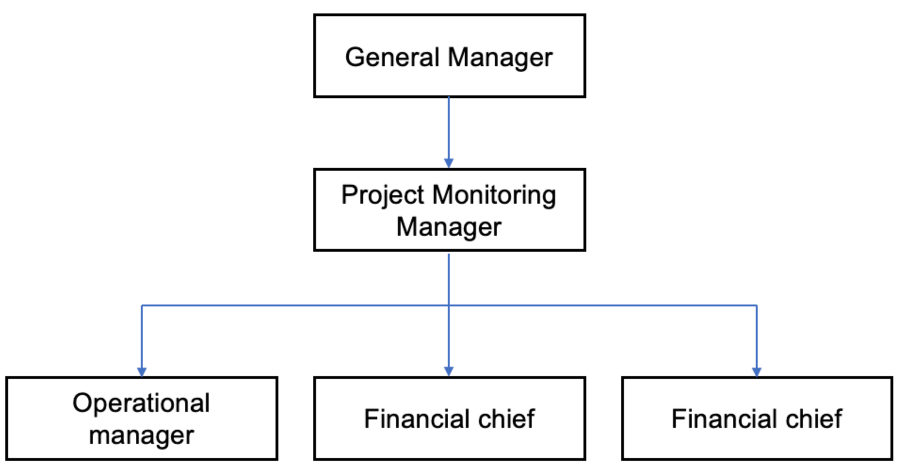
The project adopts the approved consolidated monitoring methodology ACM0002 "Grid-connected electricity generation from renewable sources" (version 20.0) to determine the emission reductions from the net electricity generation of the solar power plant. The monitoring plan is developed in accordance with the methodological requirement and will be implemented by the project participant. The monitoring plan describes in more details as below.

1. Management Structure

The project owner will use this document as guideline in monitoring of the project emission reduction performance and will adhere to the guidelines set out in this monitoring plan to ensure that the monitoring is credible, transparent and conservative.

The responsibilities of the monitoring team are as follow:

- General Manager: To be responsible for supervising the whole monitoring procedure.
- Project Monitoring Manager: To be responsible for data management and compiling monitoring report.
- Operational manager: To be responsible for collecting data, cross-checking and do internal audit.
- Financial chief: To be responsible for collection of sales receipts.
- Technical chief: To be responsible for preparing operational reports of the project activity, recording the daily operation of the solar power plant, including operating periods, equipment defects, etc.



2. Monitoring Program and Equipment

The following monitoring equipments are installed to ensure the implement of the monitoring for the project. The monitoring meters (M1 and M2) are installed at the 132kV side of the unit transformer within the project site. Usually, the main meter (hereafter referred as the meter M1) will be used to monitor the grid-connected electricity supplied to the grid by the project, and when the solar power plant is not in operation and electricity purchased from the grid will be used to ensure the minimum requirement of running the solar power plant, the meter can also be used to monitor the electricity purchased from the grid. M2 refers to the back-up meter. At the same time, the data can be monitored and recorded at the on-site control centre using a computer system. The meter reading will be readily accessible for Verra.

Before the project is in operation, both the project owner and the grid company will check the equipments to ensure their work properly. The accuracy of the above meters is no lower than 0.5.

3. Data Collection

The verification will use the main meter's data as long as the error of the meter is within the permissible tolerance. The main procedures are as follows:

I. The project owner will record the power supplied to the grid and purchased from the grid at 24:00 o'clock of last day of each month.

II. The project owner will calculate the net electricity supplied to the grid based on the records above;

III. The project owner keeps and safe keeps the records of the main meter's data readings for verification by the VVB.

If the error of the main meter exceeds the allowable tolerance or its malfunction occurs, the grid-connected electricity generated by the project will be resolved by following measures:

I. Adopting the backup meter's data (after taking the line losses into consideration), unless a test by either party reveals it is inaccuracy;

II. If the inaccuracy of the backup meter is not within the acceptable limits or it cannot work properly, the project owner and the grid company shall jointly prepare a new agreement of correct reading;

III. If the project owner and the grid company fail to reach an agreement concerning the correct reading, this matter will be submitted for arbitration according to agreed procedures.

4. Calibration of Meters & Metering

The metering equipment will be properly calibrated and checked annually according to applicable regulations. The project owner will prepare backup procedures to deal with any errors occurred to the meters. The calibration records carried out by the grid company should be provided to the project owner, and these records will be maintained by the project owner and the third party designated.

Meters should be tested by a qualified metric organization co-authorized by the project owner and the grid company within 10 days after:

I. The detection of the reading difference between the main meter and the backup meter that exceeds the allowable tolerance.

II. The equipments malfunction caused by improper operation.

All the calibration test records should be maintained safely for the verification.

5. Data Management System

To keep safely the record of the data collected during monitoring, this project will set up a complete data management system. The project will perfect the whole monitoring procedure by developing the VCS manual, tracking information from the primary source to the end-data calculations in paper document format. It is the responsibility of the project owner to provide additional necessary data and information for validation and verification requirements of respective DOE. Physical documentation such as paper-based maps, diagrams and environmental assessment will be collated in a central place, together with this monitoring plan. All paper-based information will be stored by the project owner and kept at least one copy.

At the end of each month, the monitoring data will be filed in a spreadsheet, and the paper-based printout should be also archived. Furthermore, the project owner collects the sales receipts for the electricity supplied to the grid as a cross-check, and compiled the monitoring report including the monitoring data and relevant evidence at the end of each crediting year.

All the data will be kept for two years following the end of the last crediting period.

6. Monitoring Report

After the Project Monitoring Manager collects and sorts the monitored data, the monitoring report is prepared by the designated project participate. The project developer shall make sure that the format and content of the monitoring report are consistent with the monitoring template in the registered PDD.

7. Quality Assurance and Quality Control

The workers are trained to be competent and the metering equipments are calibrated and sealed as per the industry practices at regular intervals, with the purpose to provide credible, accurate, transparent and conservative monitoring data and ensure the real, measurable, long-term GHG emission reduction from this project.

Monthly net on-grid electricity supplied data will be approved and signed off by the Manager before it is accepted and stored. This audit will check compliance with monitoring procedures in this monitoring plan. This internal audit will also identify potential improvements to procedures to improve monitoring and reporting in future years. The purpose of training is to assure those staffs are competent to conduct the monitoring plan, thus to make the monitored data accurate.

8. Training

The project owner will entrust the professional engineers and experts to train all the relative staffs before operation of generators. The training contains VCS knowledge, operational regulations, quality control (QC) standard flow, data monitoring requirements and data management regulations etc.

6 ACHIEVED GHG EMISSION REDUCTIONS AND REMOVALS

6.1 Data and Parameters Monitored

Data / Parameter	$EG_{facility,y}$
Data unit	MWh
Description	Quantity of net electricity generation supplied by the project to the Grid in year y.
Value applied:	*171,859.680* During this monitoring period, the monthly monitored data is listed as below:

Period	$EG_{export,y}$	$EG_{import,y}$	$EG_{facility,y}$
04/11/2020~30/11/2020	8,309.568	38.592	*8,270.976*
01/12/2020~31/12/2020	6,198.144	45.600	*6,152.544*
01/01/2021~31/01/2021	4,629.792	44.160	*4,585.632*
01/02/2021~28/02/2021	7,374.336	40.368	*7,333.968*
01/03/2021~31/03/2021	8,721.168	39.984	*8,681.184*
01/04/2021~30/04/2021	10,029.312	35.808	*9,993.504*
01/05/2021~31/05/2021	9,117.840	36.960	*9,080.880*
01/06/2021~30/06/2021	7,374.480	33.456	*7,341.024*
01/07/2021~31/07/2021	7,604.016	35.328	*7,568.688*
01/01/2021~31/08/2021	6,913.392	36.000	*6,877.392*

	01/09/2021~30/09/2021	9,201.600	35.904	*9,165.696*
	01/10/2021~31/10/2021	8,966.736	41.568	*8,925.168*
	01/11/2021~30/11/2021	8,721.216	44.976	*8,676.240*
	01/12/2021~31/12/2021	7,179.216	48.768	*7,130.448*
	01/01/2022~31/01/2022	5,651.232	41.616	*5,609.616*
	01/02/2022~28/02/2022	7,725.504	40.848	*7,684.656*
	01/03/2022~31/03/2022	9,353.232	40.224	*9,313.008*
	01/04/2022~30/04/2022	7,827.552	35.040	*7,792.512*
	01/05/2022~31/05/2022	8,374.992	38.352	*8,336.640*
	01/06/2022~30/06/2022	5,650.992	37.104	*5,613.888*
	01/07/2022~31/07/2022	8,741.424	38.736	*8,702.688*
	01/08/2022~31/08/2022	9,063.120	39.792	*9,023.328*

Comments

In this monitoring period, the electricity was monitored continuously and aggregated monthly.

The key information of the Main Meter and backup Meter installed at the 132kV side of the unit transformer within the project site is shown as below:

Monitoring equipment	Main meter	Backup meter
Type	ZMD402CT	ZMD402CT
Serial No	38992612	38992613
Accuracy	0.2S	0.2S
Calibration date	30/10/2020 13/10/2021	30/10/2020 13/10/2021
Validity date	29/10/2021 12/10/2022	29/10/2021 12/10/2022

The electricity data has been cross checked with sales receipts.

6.2 Baseline Emissions

The baseline emissions are the product of net electricity generation supplied to the grid expressed in MWh by the project multiplied by an emission factor.

$$BE_y = EG_{PJ,y} \times EG_{grid,CM,y}$$

Where:

BE_y	= Baseline emission in year y (tCO_2/yr)
$EG_{PJ,y}$	= Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the VCS project activity in year y (MWh/yr)
$EF_{grid,CM,y}$	= Combined margin CO_2 emission factor for grid connected power generation in year y calculated using the latest version of the "Tool to calculate the emission factor for an electricity system"

$EG_{PJ,y} = EG_{facility,y}$=171,859.680 MWh

EGy=*171,859.680*×0.5818=99,987 tCO_2e

6.3 Project Emissions

The project is a solar farm, belongs to renewable energy activity, hence PE_y =0 tCO_2e/MWh.

6.4 Leakage

No leakage is considered for this project.

6.5 Net GHG Emission Reductions and Removals

As described in the section above, emission reductions are calculated as follows:

$$ER_y = BE_y - PE_y$$

Where:

ER_y = Emission reductions in year y (tCO_2)

BE_y = Baseline Emissions in year y (tCO_2)

PE_y = Project emissions in year y (tCO_2)

The calculated ERs during this monitoring period are listed in the following table:

Year	Baseline emissions or removals (tCO_2e)	Project emissions or removals (tCO_2e)	Leakage emissions (tCO_2e)	Net GHG emission reductions or removals (tCO_2e)
04/11/2020-31/12/2020	8,391	0	0	8,391
01/01/2021-31/12/2021	55,480	0	0	55,480

01/01/2022-31/08/2022	36,116	0	0	36,116
Total	**99,987**	0	0	**99,987**

Detailed calculation has been provided in ER sheet.

This monitoring period is from 04/11/2020 to 31/08/2022 (Both days included), with totally 666 days. Based on the annual estimated emission reductions, the amount of emission reductions corresponding to this monitoring period would be 102,107 tCO_2e. The actual emission reductions in this monitoring period are 99,987 tCO_2e, which is 2.08% lower than the estimation and at the normal fluctuation range. Thus, net GHG emission reductions during this monitoring period are considered to be reasonable.